

Decoding Car Insurance: Exploring Predictors of Annual Premiums

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1. Introduction

In this study, our aim is to delve into the complex interplay of demographic factors, driving history, and policy-related characteristics on vehicle insurance Annual Premiums. We specifically focus on understanding if these factors hold the key to predicting customer responsiveness to insurance policies. Our hypothesis suggests that variables such as age, gender, driving license status, region code, previous insurance status, vehicle age, vehicle damage, policy sales channel, and vintage (customer association duration) collectively shape the annual premium. Through meticulous analysis and robust statistical modelling, we aspire to uncover the nuanced relationships between these variables and premium variability.

Central to our investigation is the fundamental question: What precise role do these diverse factors play in determining annual premiums, and can they be harnessed to predict customer responsiveness? By bridging the theoretical and practical realms of insurance industry dynamics, we aim to provide actionable insights that empower insurance providers. Our methodology integrates advanced regression techniques with a keen focus on customer responsiveness, equipping industry stakeholders with the knowledge and tools necessary to optimize pricing strategies and enhance customer engagement.

As we navigate the intricate landscape of insurance pricing, our research endeavors to offer more than just theoretical insights. We seek to empower insurance companies with actionable intelligence, enabling them to confidently navigate the ever-evolving industry landscape. Through a nuanced understanding of the factors driving annual premiums and their implications for customer behavior, we aim to pave the way for personalized, data-driven approaches to insurance pricing and customer relationship management.

In addition to examining the relationship between demographic factors, driving history, and policy-related characteristics on vehicle insurance premiums, we will also explore the impact of market dynamics and regulatory influences. Understanding how external factors such as market competition, regulatory changes, and economic conditions interact with internal variables can provide deeper insights into premium variability. By incorporating these broader contextual factors into our analysis, we aim to offer a comprehensive understanding of the forces shaping the insurance industry landscape.

Furthermore, our study will not only focus on predictive modeling but also on assessing the fairness and equity of insurance pricing practices. We recognize the importance of ensuring that insurance premiums are not only predictive but also fair and unbiased across different demographic groups. Through careful examination and analysis, we will evaluate the extent to which current pricing practices align with principles of fairness and identify areas for improvement. By promoting transparency and equity in insurance pricing, our research aims to contribute to a more inclusive and socially responsible insurance industry.

2. Literature Review

The study "Decoding Car Insurance: Exploring Predictors of Annual Premiums and Customer Responsiveness" seeks to investigate the influence of demographic factors on annual premiums for vehicle insurance and to predict customer responsiveness. In this literature review, we explore key articles that provide insights into the factors affecting car insurance premiums, aligning with the project's objectives.

1. Age and Gender Impact on Car Insurance Rates:

- **Source:** Jason Metz's analysis on Forbes.
- Metz's findings highlight the substantial impact of age and gender on car insurance rates, supporting the project's hypothesis.
- Young drivers face higher premiums due to limited driving experience, and a noticeable decline begins at age 25. Gender differences are evident, with men generally paying more than women.
- This analysis contributes valuable context to the project, aligning with the focus on demographic factors and setting the stage for a nuanced exploration of predictors.

2. Understanding Insurance Premiums and Factors Influencing Them:

- **Source:** Julia Kagan's article on Investopedia.
- Kagan's article provides a comprehensive overview of insurance premiums, emphasizing factors such as coverage type, age, location, claims history, and risk considerations.
- The article delves into how premiums are calculated, considering the role of actuaries and the impact of advanced algorithms.
- This foundational knowledge enhances the project's contextualization of demographic and policy-related predictors in the realm of insurance premiums.

3. Why Do Inner City Residents Pay Higher Premiums? The Determinants of Automobile Insurance Premiums:

- **Source:** Study by Paul M. Ong and Michael A. Stoll from UCLA School of Public Affairs.
- This study investigates disparities in auto insurance premiums in poor and minority neighborhoods, considering both risk factors and redlining (socioeconomic) factors.
- It provides insights into how socioeconomic factors remain statistically significant even after accounting for risk factors, offering a nuanced understanding of premium variations.
- The study's findings contribute to the project's exploration of predictors and highlight the interplay between socioeconomic factors and insurance premiums.

4. How Age Affects Car Insurance Prices for Men and Women:

- **Source:** Lindsay Bishop's study on ValuePenguin.

- Bishop's study explores the impact of age and gender on car insurance prices, providing key insights into age-related pricing differences and gender-based variations.
- The study directly aligns with the project's focus on exploring predictors of annual premiums, particularly age and gender.
- The findings offer valuable information on the influence of demographic factors on car insurance prices, enriching the project's analytical framework.

5. **Good drivers pay less: A study of usage-based vehicle insurance models:**

- **Source:** Study by Yiyang Bian et al. on ScienceDirect.
- This paper introduces the concept of Usage-Based Insurance (UBI) and a Behavior-centric Vehicle Insurance Pricing (BVIP) model, emphasizing the impact of behavioral data on driving risk.
- The study's empirical data analysis showcases the effectiveness of BVIP in risk-level classification and proposes innovative approaches to personalized UBI pricing strategies.
- Insights from this study contribute to the project's exploration of predictors, offering a perspective on the evolving landscape of insurance through innovative pricing models.

6. **Conducting customer behaviour analysis in insurance and its benefits:**

- **Source:** Netscribes article.
- The article explores the significance of customer behavior analysis in the insurance industry, emphasizing personalized experiences, increased conversion rates, and spotting market trends.
- It introduces concepts such as customer segmentation, predictive analysis, and behavior tracking mechanisms, offering insights into the role of data analytics in understanding customer behavior.
- The article's focus on customer-centric models aligns with the project's objective of predicting customer responsiveness.

7. **An analysis of the risk factors determining motor insurance premium in a small island state: the case of Malta:**

- **Source:** Study by Ayrton Magri, Andre Farrugia, Francis Valletta, Simon Grima in the International Journal of Finance, Insurance and Risk Management.
- The study from Malta investigates risk factors influencing motor insurance premiums, advocating for the use of penalty points and telematics for better risk assessment.
- Findings suggest that Maltese insurers may be falling short in risk identification, impacting premium accuracy.
- The study provides valuable insights into risk factors affecting motor insurance premiums, aligning with the project's focus on predictors of annual premiums.

In conclusion, these literature reviews provide a comprehensive background for the project, offering insights into various factors influencing car insurance premiums, including demographic, behavioural, and risk-related elements. The integration of these insights into the multiple regression analysis will contribute to a more thorough understanding of predictors and their collective impact on annual premiums and customer responsiveness.

3. DATA, DATA SOURCES, AND DATA CHARACTERISTICS

The data utilized in this study was collected through the Kaggle website (<https://www.kaggle.com/code/shardulmehetar/vehicle-insurance-eda-modeling-top-6/input?select=test.csv>), and while the cleaning and preprocessing stages had been previously executed, we conducted a thorough examination to ensure data integrity. This involved meticulous checks for missing values, scrutiny of anomalous patterns or outliers, and validation of measurement units. Appropriate imputation techniques were applied to handle missing values, and outliers were identified and addressed through methods such as trimming or transformation. Furthermore, standardization and normalization techniques were employed to bring features to a common scale, enhancing comparability. Additionally, we crafted a robust data dictionary to provide a comprehensive reference for each variable in the dataset, including variable names, units of measurement, conversion details, and the level of granularity. This approach not only upholds the integrity of the data but also establishes a solid foundation for subsequent analyses and interpretations, ensuring transparency and aiding others in better understanding the intricacies of the dataset.

Variable	Description	Data Type	Possible Values
id	Unique identifier for each policyholder	Numeric	Unique numerical identifiers
Gender	Gender of the policyholder	Factor	Male, Female
Age	Age of the policyholder	Numeric	Integer values representing age
Driving License	Whether the policyholder has a driving license	Numeric	0 (No), 1 (Yes)
Region Code	Code representing the region of the policyholder	Numeric	Integer codes for different regions

Previously Insured	Whether the policyholder was previously insured	Numeric	0 (No), 1 (Yes)
Vehicle Age	Age of the vehicle	Factor	< 1 Year, 1-2 Year, > 2Years
Vehicle Damage	Whether the vehicle has been damaged in the past	Factor	No, yes
Annual Premium	Annual premium amount for insurance	Numeric	Continuous numerical values
Policy Sales Channel	Channel code for policy sales	Numeric	Integer codes for different sales channels
Vintage	Number of days since the policyholder was associated	Numeric	Integer values representing the number of days

SUMMARY STATISTICS:

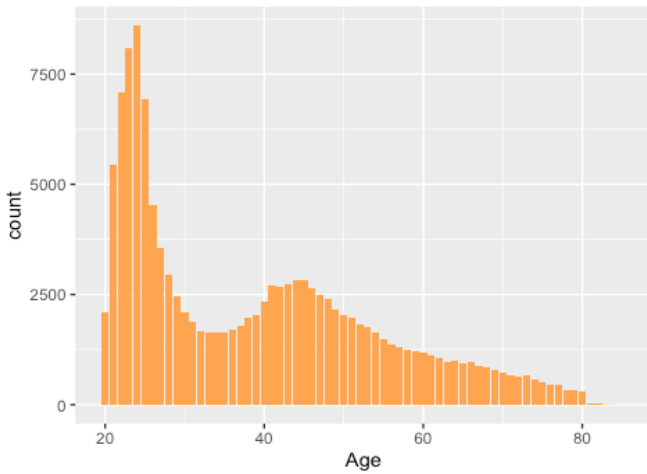
Variable	Min	Median	Mean	Max	Count
Age	20	36	38.77	85	127037
Annual_premium	2.630	31.462	30.525	472.042	127037
Vintage	10	124	160	299	127037

Variable	Male	Female
Gender	58801	68236
Vehicle_Age	55019	66699
Vehicle_Damage	63202	63835
Previously_Insured	68595	58442

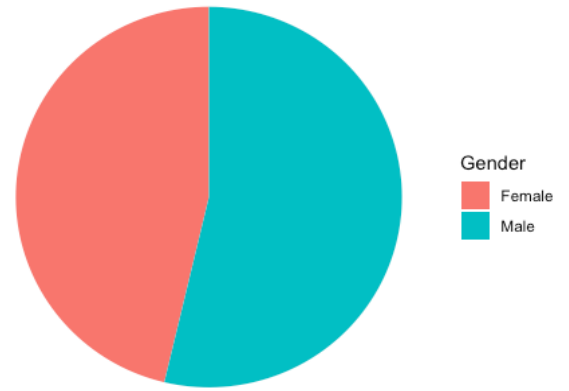
Significant disparity between the minimum and maximum values suggests a greater likelihood of outliers or inaccuracies in the data

Data Visualization:

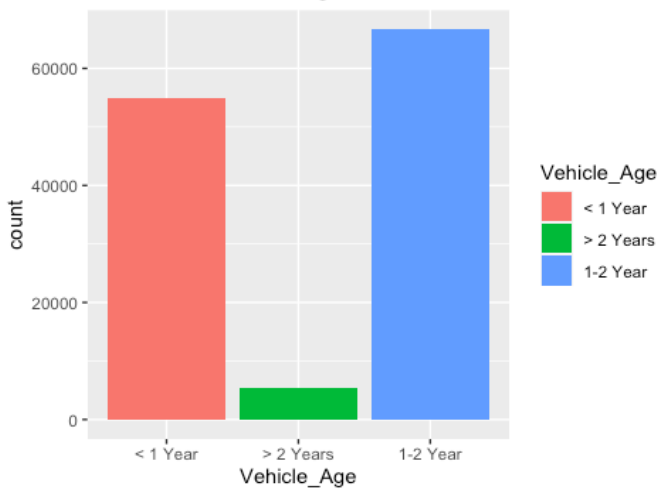
Bar Plot of Age Distribution



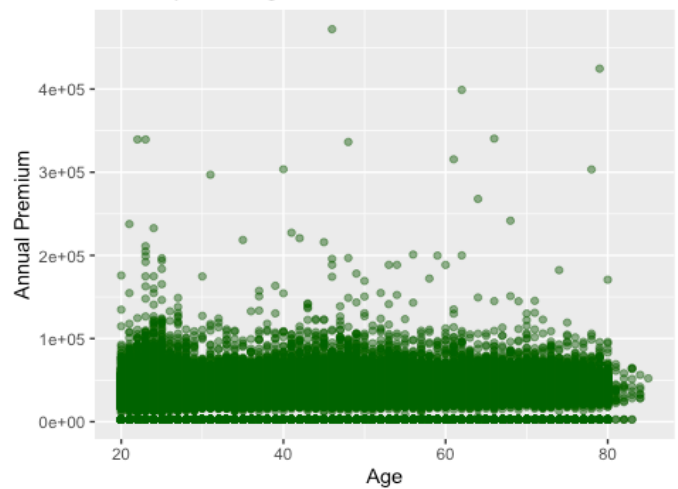
Pie Chart of Gender Distribution



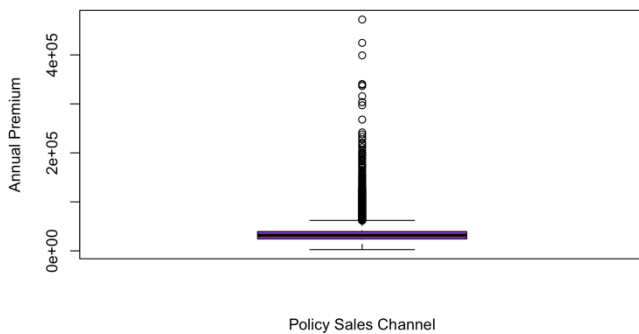
Bar Plot of Vehicle Age Distribution



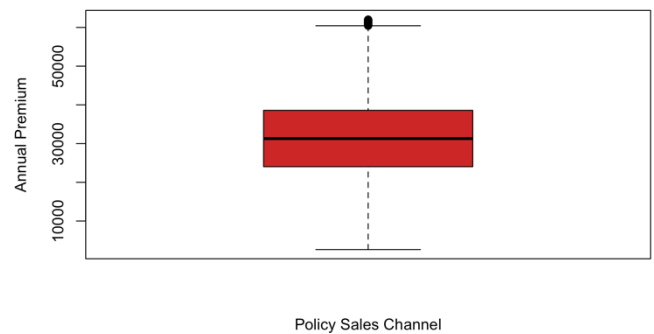
Scatterplot of Age vs. Annual Premium



Annual Premium by Policy Sales Channel Before Removing Outliers



Annual Premium by Policy Sales Channel After Removing Outliers



DATA DIMENSION REDUCTION:

In this project, the dataset spans from 2018 to 2019 and includes insurance-related data comprising 127,487 initial observations. These observations are characterized by 11 predictor variables, such as id, Gender, Age, and others, with Annual Premium serving as the outcome variable. After data pre-processing, the dataset was refined to 123,569 observations, maintaining the same predictor and outcome variables. Because only 227 people in this dataset do not have a driving license out of more than 123,000 observations, we could consider skipping the driving license variable and eliminating the corresponding rows. The decision to reduce the dataset's dimensionality aimed to streamline our analysis and enhance model efficiency. By focusing on significant predictors and removing outliers, we aimed to simplify interpretation while ensuring our analysis remains focused on the factors impacting Annual Premium. Reducing noise in the dataset was another key motivation. Outliers can introduce variability and distortions, potentially leading to inaccuracies. By removing outliers, we aimed to create a cleaner dataset that better captures underlying patterns and relationships, improving the reliability and interpretability of our findings. Additionally, prioritizing key features with the most significant impact on Annual Premium allows for a more tailored model. This approach filters out noise and irrelevant information, aiming to develop a more efficient and effective predictive model, providing valuable insights into insurance pricing dynamics.

4. Methodology

1) Introduction:

Our study aims to explore the factors influencing annual car insurance premiums through statistical modelling techniques. After reviewing relevant literature, we opted for multiple regression analysis as our primary model. This choice aligns with our research question, which seeks to understand how demographic factors, driving history, and policy-related characteristics collectively impact insurance premiums. Multiple regression analysis allows us to evaluate the combined influence of these factors on premium variability.

We selected multiple regression analysis over alternative models due to its capability to handle multiple predictor variables and capture complex relationships with the dependent variable (annual premium). Moreover, its interpretability and versatility are well-documented in both academic research and industry standards.

2) Approach and Methodology:

Model Selection: Multiple regression analysis was chosen for its capacity to handle multiple predictor variables and assess complex relationships with the annual premium.

Data Pre-processing: We employed outlier detection using the interquartile range (IQR) method and subsequent data cleaning to enhance the robustness of our analysis. Additionally, we transformed categorical variables like 'Region Code' and 'Previously Insured' into factors to integrate them properly into the regression analysis.

Data Visualization: Before regression analysis, we conducted exploratory data analysis (EDA) to understand variable distributions, relationships, and potential patterns using histograms, scatter plots, and residue plots.

Justification for Chosen Model:

- **Compatibility with Data:** Multiple regression analysis is well-suited for our dataset, containing both numerical and categorical predictor variables.
- **Complexity of Problem:** Given the multifaceted nature of predicting car insurance premiums, multiple regression analysis can effectively handle various predictors simultaneously.
- **Availability of Resources:** Widely used in statistical practice, multiple regression analysis offers readily available resources for implementation, ensuring feasibility with our available resources.

Advantages of Chosen Model:

- **Interpretability:** Multiple regression analysis provides interpretable coefficients, facilitating understanding of predictor variable effects on annual premiums.
- **Flexibility:** The model accommodates both numerical and categorical predictors, enabling analysis of diverse data types.

5. Empirical Results

We conducted a comprehensive evaluation of Model 1, Model 2, and Model 3 - comparing their adjusted R-squared values, residual standard errors, significance of predictors, and overall interpretability. Both models yielded identical results, but Model 2, excluding the non-significant predictor 'Vintage,' was preferred for its simplicity and consistency in predicting annual premiums.

Model 1:

- **Variables Included:** id, Age, Gender, Region_Code, Driving License, Previously_Insured, Vehicle_Age, Vehicle_Damage, Policy Sales Channel, Vintage
- **Adjusted R-squared:** 0.2897
- **Residual Standard Error:** 12430

- **Significant Predictors:** Age, Gender(Male), Region_Code, Previously_Insured, Vehicle_Age, Vehicle_Damage

Interpretation: The regression results from Model 1 reveal that older age, being male, specific region codes, lack of previous insurance, having a vehicle age greater than 2 years, and experiencing vehicle damage are all positively associated with higher annual premiums. Notably, certain predictors such as driving license demonstrate no significant impact on premium variability likely due to its universal presence among individuals in the dataset.. The model effectively explains a substantial portion of the variability in annual premiums, with a multiple R-squared value of 0.2909, indicating a robust fit to the data.

Model 2:

- **Variables Included:** Age, Gender, Region_Code, Previously_Insured, Vehicle_Age, Vehicle_Damage
- **Adjusted R-squared:** 0.2099
- **Residual Standard Error:** 13110
- **Significant Predictors:** Age, Gender, Region_Code(Specific), Previously_Insured, Vehicle_Age, Vehicle_Damage
- **Interpretation:** In Model 2, the analysis underscores the significance of age, male gender, particular region codes, lack of previous insurance, having a vehicle age greater than 2 years, and experiencing vehicle damage as predictors of higher annual premiums. The absence of driving license as a predictor reflects its universal presence among individuals in the dataset, hence it does not significantly influence premium variability. The model explains approximately 21.02% of the variability in annual premiums, suggesting a notable portion of the variation is captured by the included predictors.

Model 3:

- **Variables Included:** Age, Gender, Region_Code, Previously_Insured, Vehicle_Age, Vehicle_Damage, interaction terms of Age and Vehicle_Age, interaction terms of Previously_Insured and Vehicle_Damage
- **Adjusted R-squared:** 0.2195

- **Residual Standard Error:** 13030
- **Significant Predictors:** Age, Gender, Region_Code(Specific), Previously_Insured, Vehicle_Age, Vehicle_Damage, Age:Vehicle_Age, Previously_Insured:Vehicle_Damage
- **Interpretation:** Based on Model 3, the regression analysis highlights several factors influencing annual premiums. Specifically, older age, male gender, certain region codes, lack of previous insurance, having a vehicle age greater than 2 years, and experiencing vehicle damage are all associated with higher premiums. Additionally, interaction terms such as age with vehicle age and previously insured with vehicle damage also play significant roles in premium determination. Overall, this model explains approximately 21.99% of the variability in annual premiums, suggesting a substantial portion of the variation is accounted for by the included predictors and interactions.

Advantages of Model 3:

- **Enhanced Predictive Power:** Model 3 incorporates interaction terms, allowing for a more nuanced understanding of how certain variables jointly influence annual premiums. This increased complexity may result in improved predictive accuracy compared to Models 1 & 2.
- **Captures Nonlinear Relationships:** By including interaction terms, Model 3 can capture nonlinear relationships between predictors and the outcome variable, providing a more realistic representation of the data.
- **Greater Flexibility:** The inclusion of interaction terms adds flexibility to the model, allowing for more tailored predictions based on different combinations of predictor variables.

VIF (Variance Inflation Factor) on Model 3:

	GVIF	Df	$GVIF^{1/(2*Df)}$
Gender	1.034875	1	1.017288
Age	2.715994	1	1.648027
Region_Code	1.258195	52	1.002211
Previously_Insured	3.165743	1	1.779253
Vehicle_Age	3.144823	2	1.331677

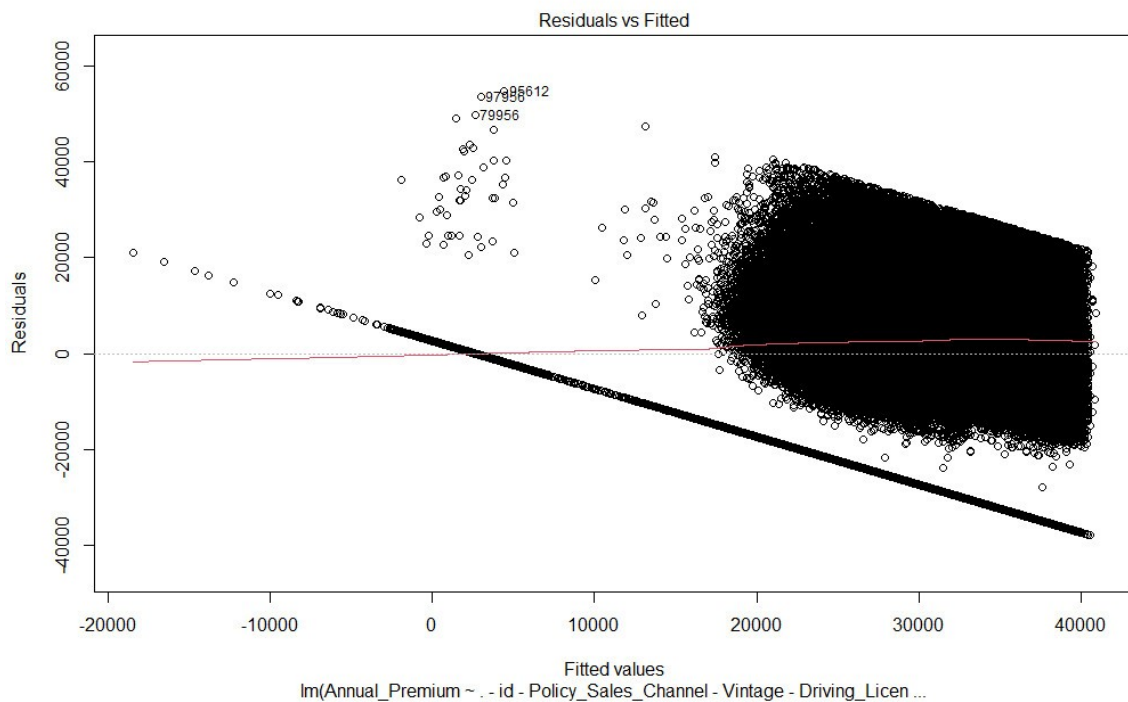
Vehicle_Damage 3.188034 1 1.785507

Interpretation: The variance inflation factor (VIF) values for the corrected Model 3 suggest varying degrees of multicollinearity among predictor variables. The variables "Gender," "Region_Code," and "Vehicle_Damage" exhibit low levels of multicollinearity, with VIF values close to 1. However, variables such as "Age," "Previously_Insured," and "Vehicle_Age" display moderate multicollinearity, as indicated by VIF values around 2 to 3. While the removal of the "Age:Vehicle_Age" interaction term has alleviated some multicollinearity concerns compared to the original Model 3, there may still be residual correlations among certain predictors. Future iterations of the model may benefit from additional refinement to further mitigate multicollinearity and enhance the model's predictive accuracy and stability.

Model	RSE	Adjusted R-squared	Explanation
Model 1	12430	0.2897	Model 1 has the lowest residual standard error, indicating that it has the smallest difference between the observed values and the values predicted by the model. Additionally, it has the highest adjusted R-squared value, suggesting that it explains the variation in the response variable (Annual Premium) better than the other models.
Model 2	13110	0.2099	Model 2 has a higher residual standard error compared to Model 1, indicating a larger difference between the observed values and the values predicted by the model. Its adjusted R-squared value is lower than Model 1, suggesting that it explains less of the variation in the response variable.
Model 3	13030	0.2195	Model 3's residual standard error is slightly higher than Model 1 but lower than Model 2. Its adjusted R-squared value falls between Model 1 and Model 2, indicating that it explains more variation in the response variable than Model 2 but less than Model 1. Model 3 incorporates interaction terms to capture nuanced relationships, potentially improving predictive accuracy

Conclusion: In summary, the comparison of Models 1, 2, and 3 highlights the iterative nature of predictive modeling. While Models 1 and 2 offer simplicity and ease of interpretation, Model 3's inclusion of interaction terms provides enhanced predictive power and flexibility. The choice of model ultimately depends on the specific goals of the analysis and the trade-offs between simplicity and predictive accuracy. By carefully considering the strengths and limitations of each model, insurance companies can make informed decisions to better understand and predict annual premiums, ultimately improving their ability to serve their customers effectively.

Model 4 (Polynomial Regression on Model 3):



Call:

```
lm(formula = Annual_Premium ~ . + I(Age^2) + sqrt(Age) - id -  
  Policy_Sales_Channel - Vintage - Driving_License + Age *  
  Vehicle_Age + Previously_Insured * Vehicle_Damage, data = data_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-37122	-4617	1803	8000	49159

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-44428.327	6869.543	-6.467	1.00e-10 ***
GenderMale	197.167	75.735	2.603	0.00923 **
Age	-2973.449	346.324	-8.586	< 2e-16 ***
Region_Code1	-724.057	854.555	-0.847	0.39683
Region_Code2	23019.184	616.798	37.320	< 2e-16 ***
Region_Code3	19855.575	555.523	35.742	< 2e-16 ***

Region_Code52	23425.383	1476.749	15.863	< 2e-16 ***
Previously_Insured	2839.006	159.216	17.831	< 2e-16 ***
Vehicle_Age> 2 Years	-26176.995	1276.495	-20.507	< 2e-16 ***
Vehicle_Age1-2 Year	-28623.571	822.503	-34.801	< 2e-16 ***
Vehicle_DamageYes	1780.021	158.732	11.214	< 2e-16 ***
I(Age^2)	5.444	1.180	4.614	3.95e-06 ***
sqrt(Age)	23651.763	2954.938	8.004	1.21e-15 ***
Age:Vehicle_Age> 2 Years	833.921	31.689	26.316	< 2e-16 ***
Age:Vehicle_Age1-2 Year	849.355	26.241	32.368	< 2e-16 ***
Previously_Insured:Vehicle_DamageYes	-5339.336	285.512	-18.701	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13030 on 123376 degrees of freedom

Multiple R-squared: 0.2217, Adjusted R-squared: 0.2213

F-statistic: 557.8 on 63 and 123376 DF, p-value: < 2.2e-16

We utilized polynomial regression to capture non-linear age-premium associations, evidenced by terms like Age^2 and $\text{sqrt}(\text{Age})$. This approach accommodates intricate age-related effects often overlooked by linear models, revealing nuanced relationships across different age groups. It underscores the sporadic nature of age as a predictor, with varying influences depending on context.

6. Conclusion

Our study offers a comprehensive examination of the intricate factors influencing annual car insurance premiums, going beyond conventional demographic considerations. Through the application of advanced regression models, we challenge simplistic assumptions, particularly regarding the influence of age on premium determination, and unearth valuable insights. By incorporating sophisticated mathematical techniques, such as polynomial terms for age, and highlighting the significance of factors like region code and vehicle age, our models provide a deeper understanding of the diverse influences shaping insurance costs.

Our research underscores the importance of utilizing sophisticated statistical methods to enhance insights into insurance pricing dynamics, benefiting both insurance providers and policyholders. Insurance companies can refine their pricing strategies by considering age variations and geographical impacts alongside vehicle age, leading to more tailored and competitive pricing structures. For policyholders, these insights empower informed decision-making, enabling them to select insurance coverage that aligns with their needs and potentially negotiate better prices.

Furthermore, our study sheds light on the multifaceted nature of factors affecting annual car insurance prices, extending beyond conventional considerations, and contributes to a deeper understanding of insurance pricing dynamics. In comparison with existing literature, our findings offer unique insights into the complex interplay of factors influencing annual premiums, integrating a broader range of predictors and providing a more holistic view of premium determination. Future research could explore additional factors influencing annual car insurance premiums, such as environmental factors, technological advancements, and regulatory changes, and conducting longitudinal studies to track changes in premium determinants over time could provide valuable insights into evolving trends within the insurance industry.

In conclusion, our research underscores the transformative potential of data-driven decision-making in the insurance sector, enabling insurance providers to enhance pricing strategies, improve customer satisfaction, and foster a more transparent and competitive insurance market.

7. References

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